

the breadboard. You can create boards of different sizes, up to 100×100 holes.

Pick and place, create and edit, on a single screen

The graphical user interface (GUI) of PerfDesign is neat and simple, designed for easy usage. Resistor, diode, transistor, capacitor, integrated circuit, trimmer, LED, potentiometer, switch and jumper—all design components including connectors, wires and track cutters are placed right at one end of the screen. At the other end, there is Explorer window that lets you play with different combinations of values and fine-tune circuits to achieve desired results, instead of changing, say, the resistor each time.

As you add each component, you can choose properties to have a basic design to begin with. The board is created right at the centre of the screen. Just below the board, there is a component overview area that lists out all components used (parts list) in the design, along with their details.

Working your way around the board is very easy. Just a few clicks of the mouse lets you add new components, drag these around to create the intended design, resize these, change properties and basically create, re-create or tune the board to achieve perfection. You can also zoom in to ensure proper connectivity, or pan the entire design to one end in order to re-structure the board. If the board is too huge and complicated, use

Component Explorer menu to get a compact view, with components listed, sorted by type and name.

The board is designed, what next

Overwhelmed by the number of easy-to-design PCB tools, we rarely use perf or strip boards anymore. But the fact remains that such boards are probably the best choices for quick prototypes and, if well done, long lasting ones, too. When compared to printed boards, these are certainly easier to work with and, more importantly, to debug. Yes, you might end up putting the capacitor leads into wrong holes, or solder something wrong, but redoing it will not take too much effort.

You have created your design and laid it out neatly on the perf or strip board. Now, all you need to do is copy the same on the real board. This is a cheap solution that lets you check your circuit's functions instantly and modify these in a short span. Just make sure to get the soldering techniques right, remember to scrap off unnecessary connections if working with a strip board, and you have a nice design on the front side and a neat layout at the back of the board.

A project for every board

While working with PerfDesign, remember to save each design as a new project. Each time you save one, the files get stored in binary format in dat extension. As you create your circuit, you can take snapshots that can then directly go into reports. PerfDesign stores such images in jpeg format.

PerfDesign has been completely created using C# and released under GNU GPL licence. There are a few other tools that let you create designs on strip boards like Stripboard Designer, VeeCAD, BlackBoard Circuit Designer, VeroDes and LochMaster, to name a few. **EFY**

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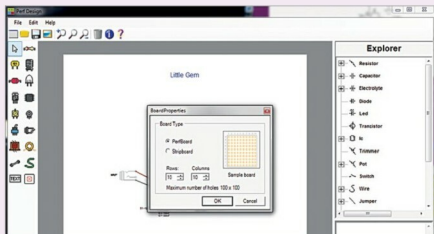


Fig. 2: Customise your board

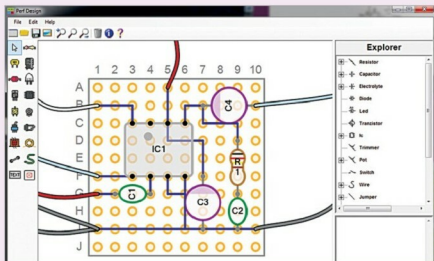


Fig. 3: Designing on a perf board